Southern New Hampshire University

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design and develop a system for DriverPass that allows customers to prepare for their driving tests through online classes, practice exams, and on-the-road training. The client, DriverPass, wants a system that manages user accounts, schedules driving lessons, tracks progress, and provides access to training materials both online and through downloadable reports.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass wants to solve the problem of high failure rates on DMV driving tests by providing better training resources. The system should allow users to take online courses, complete practice tests, and schedule in-person driving lessons. Key components include a web-based platform, a scheduling system, user account management, reporting tools, DMV content integration, and a database to store customer, instructor, and appointment information.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

· The system should allow customers to register, log in, and manage their accounts
· The system should allow users to schedule, modify, and cancel driving appointments
· The system should provide online classes and practice tests aligned with DMV standards
· The system should track user activity and generate reports
· The system should allow administrators to manage users, packages, and system data
· The system should ensure secure handling of personal and payment information

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Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system will be web-based and accessible through browsers on desktop and mobile devices
- The system should load pages within 2–3 seconds under normal usage
- The system should support multiple users simultaneously without performance issues
- The system should be updated regularly to reflect DMV changes and system improvements

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should run on common operating systems such as Windows, macOS, and mobile platforms
- The system will require a backend database to store user, appointment, and system data
- The system should be cloud-based to reduce maintenance and improve accessibility

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Each user will have a unique login to distinguish accounts
- The system should validate all input fields (e.g., required fields, correct formats)
- The system should notify administrators when errors or inconsistencies occur
- The system should maintain accurate tracking of reservations and user activity

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should allow administrators to add, remove, or modify users without changing code
- The system should allow enabling or disabling packages as needed
- The system should be flexible enough to support future updates or new features
- IT administrators should have full access to manage and maintain the system

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?

· Users must log in with a username and password to access the system
· Data should be encrypted during transmission (HTTPS)
· Sensitive data such as payment information must be securely stored
· Accounts should be locked after multiple failed login attempts
· Users should be able to reset their password if forgotten
· Role-based access control should restrict access based on user type

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

· The system shall allow users to create and manage accounts
· The system shall allow users to log in and log out securely
· The system shall allow customers to schedule driving lessons
· The system shall allow customers to cancel or modify reservations
· The system shall assign drivers and vehicles to scheduled lessons
· The system shall provide online training materials and practice tests
· The system shall track user activity and maintain logs of changes
· The system shall generate downloadable reports for users and administrators
· The system shall allow administrators to manage users and system data
· The system shall integrate with DMV updates and notify users of changes

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

· The system should be accessible through a web browser on desktop and mobile devices
· Customers should be able to register, schedule lessons, and access training materials
· Administrators should be able to manage users, appointments, and system settings
· The interface should be simple, user-friendly, and easy to navigate
· Users should interact with the system through forms, dashboards, and scheduling tools

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

· Users have access to the internet and a compatible device
· Users have basic knowledge of how to use web applications
· The DMV will provide updated rules and content for integration
· Payment processing systems will function properly with the platform

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

· The system may depend on internet connectivity for full functionality
· Initial system features may be limited due to time and budget constraints
· Future customization of packages may require additional development
· Integration with DMV systems may be limited by external factors

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

[Insert chart]